Aryan Padmakar Deshmukh

Software Engineer / Go / Ruby on Rails / React / AWS

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Summary

Software engineer building and running a production IoT metering and utility-billing platform in **Go**, **Ruby on Rails** and **React** on **AWS**. I own work end to end: billing, site onboarding, MQTT ingestion, admin UIs and releases. Shipped **30+ pull requests** across 4 repositories in 9 months; converted from intern to full-time in 6.

Experience

Josh Software *IoT water-metering & utility-billing platform*

Pune, India | Jan 2026 – Present

Associate Software Engineer — Jul 2026 – Present

- **Billing & payments:** owned the billing module across Rails and React: built per-site platform fees, offline bill payments, fixed or rolling due dates and approval reminders; corrected **129 of 140** wrong due dates and billed usage above the top slab.
- **Site onboarding:** built onboarding end to end: company and site setup, a flats import (presigned S3, Sidekiq) and bulk resident mapping; onboarded a **195-meter** block with 0 rows refused and provisioned 480 devices.
- **Go/MQTT ingestion:** built site-level data-collection control (Go ingestion drops readings for inactive sites, toggled from React, one query path across **10** call sites) and DB-driven sensor config so new clients need no YAML edit.
- **Imports & reporting:** shipped a meter-remark bulk import making one device-service call per upload instead of per row, with a 409 concurrency guard, and a per-meter hourly usage breakdown with Excel/CSV export.
- **Infrastructure:** built and run staging for the Go/MQTT service on AWS EC2 (Docker, systemd, nginx), now on a reproducible Docker build; diagnosed Postgres exhaustion (**100/100**) and a 408-job Sidekiq backlog.
- **Releases & reviews:** run production releases across 4 repositories (8 PRs in one cut, no regressions) and review payment and background-job PRs, catching concurrency and data-loss issues before merge.

Software Engineer Intern — Jan 2026 – Jun 2026

- **Go MQTT pipeline:** reworked event ordering in the Go ingestion pipeline (QoS 0, device timestamps) after tracing negative water readings to QoS 2 redelivery; those support tickets **stopped**.
- **Meter health:** built a Go cron job emailing weekly non-responsive-meter reports through AWS SES, and Rails daily reports bucketing unreachable meters by 24 h and 3+ days.
- **Billing & payments:** built a site-configurable penalty component replacing duplicated logic, and integrated a bank payment gateway in React with live balance and limit checks.
- **Consumption tools:** built an admin consumption-correction API and an hour-wise usage report with Excel/CSV export.

InveeSync *Frontend Developer Intern*

Remote | Dec 2024 – Jan 2025

- React/Redux workflows for a manufacturing-automation SaaS, with client-side caching to cut redundant API calls.

Projects

Smart Rule-Based Approval Engine *Go, Gin, PostgreSQL, React, TypeScript* — Live | GitHub

- Rules engine auto-deciding leave, expense and discount requests; Go cron expiry job, JWT roles, React UI.

Fintech Fraud Detection *Ruby on Rails, PostgreSQL, Solid Queue* — Live | GitHub

- Real-time transaction risk scoring with device fingerprinting; fraud alerts and weekly summaries via Solid Queue jobs.

Technical Skills

Languages	Go, Ruby, C++, JavaScript, TypeScript, SQL
Backend & Data	Ruby on Rails, Gin, REST APIs, Sidekiq, MQTT, RSpec; PostgreSQL, TimescaleDB, MongoDB, Redis
Frontend	React, Redux-Saga, Zustand, TanStack Query, Next.js
Cloud & DevOps	AWS (EC2, S3, SES), Docker, nginx, systemd, Linux, Git, GitHub

Writing & Open Source

- **Medium:** **three** posts on production debugging, e.g. *The Water That Flowed Backwards* (MQTT ordering in Go).
- **Open source:** merged PRs to CopyQ (12k stars, C++/Qt), prometheus-operator (10k stars, Go/Kubernetes) and Kepler (1.6k stars, Go/Kubernetes); reviews with fix PRs on helm/helm (30k stars), one adopted.

Education

B.Tech, Computer Engineering (Honours in Cyber Security), KKWIEER, Nashik	**9.26**	2026
Class XII (HSC), Vikas High School, Bhandara	**91.67%**	2022
Class X (CBSE), Royal Public School, Bhandara	**93.00%**	2020

Achievements & Leadership

- **Only student selected on day one** of campus placements, by the first recruiter on campus (500+ candidates).
- Ranked **top 5** in the department; **Vice President, CSI Student Chapter** (2024–25), led an e-waste recycling drive.
- Captain of the school basketball team (district runners-up).